High-Frequency Market Making Infrastructure Six: Market Data Caliber Three

SK (@StepOneAi) · 8 August 2026 · posted on X · translated from Chinese

The first two articles discussed the two layers of funding rate calibers: the outer-layer average isn't the same function, and the premium index being averaged refers to depth rather than the order book. This article discusses how the rate has upper and lower bounds, and those values that are capped cannot be used as normal observations. This issue actually spills over from funding rates, so it's a standalone article.

A batch of identical numbers—we'd estimated the tail from the rate history, but the extreme values we got were always too small, not matching the actual feel. Later, when we looked at the raw sequence, we found a batch of values crammed at the same number, exactly identical, appearing repeatedly. That's not a coincidence; it's a cap.

First, let's clarify two terms

There are two concepts in statistics that are easy to confuse.

Censoring: For this observation, you know it exists, but you only know it falls outside a certain boundary, not the specific value. When the rate hits the upper limit, the true premium pressure might far exceed that limit, but the sequence only records the number at the limit.

Truncation: Observations outside the range don't even enter the sample; you don't even know they exist.

Rate capping belongs to censoring—the records are still there, but the values are flattened. This distinction is important because the handling methods differ, and censoring is more insidious: the data looks complete, every row has a value, no missing data, and the pipeline won't throw an error.

Hard boundaries are more common than you might think; once you realize this problem, you'll find trading data full of hard boundaries everywhere:

- Funding rate upper and lower limits
- Matching price protection bands, price limit ranges
- Transaction prices generated when various price protection mechanisms trigger
- Precision truncation in data fields, where parts smaller than the minimum tick size are wiped flat

- Depth interfaces only push a fixed number of levels; deeper parts aren't absent—they just aren't given to you

The last one is especially worth noting. The total order volume you calculate from a fixed number of levels is essentially an observation censored at the last level. How deep the market is—this number can't answer that.

If you use it directly, the tail gets flattened. The extreme intervals of the distribution are artificially shifted to the boundaries, so the estimated extreme quantiles and tail indices are all biased optimistic. This is the first one we ran into.

Volatility is biased low. The boundaries chop off a chunk of variance; estimating volatility from a censored sequence will be systematically too small. And the degree of bias is largest in extreme markets—exactly when you need it to be accurate.

Regression coefficients are biased. If a censored variable is the dependent variable in a regression, the OLS coefficients are biased, and the direction of bias depends on the censoring proportion. When the censored sample proportion isn't high, it's easy to overlook, but extreme-period samples are scarce to begin with, so censoring leaves almost nothing.

Thresholds move—if the censoring threshold is a fixed constant, handling it is fairly straightforward. But in the scenarios we've encountered, the threshold itself is adjustable. For example, designs that tie the rate cap to the maintenance margin rate, roughly in the form of `Capped F = clamp(F, −k × maintenance margin rate, +k × maintenance margin rate)`. The maintenance margin rate comes from the margin tiers, k is a multiple set per contract, and both can be adjusted. So the censoring point isn't a single number—it's a curve that moves over time.

This follows the same structure as the previous article: what you think is a constant is actually a variable hanging off the leverage parameters. The difference is only that in the previous one, it affects the measurement position; here, it affects the truncation position. You can't just flag which samples are censored—you also need to know what the threshold was at the time. The former can be spotted by comparing values; the latter requires parameter history.

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